SPA617 Radio Astronomy Laboratory
RF Components: Characterization and Measurement

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1 Aim

To understand the operation and experimentally characterize the different filters, mixer, power splitter and power combiner used in an RF receiver chain

2 Objective

- To identify, characterize and determine cutoff frequencies of the RF filter.
- To demonstrate a quarter-wave open-circuit stub as a notch filter.
- To observe frequency conversion in an RF mixer and estimate its conversion loss.
- To measure the power division and insertion loss of a power splitter/combiner.
- To estimate the input reflection coefficient and return loss of the filter.

3 Theory

3.1 RF components in an RF receiver chain

The RF receiver chain contains several RF components which are used to process the radio signal. The radio signal is received from the antenna, which converts EM waves into electrical signals. The amplifier is used to amplify the weak signal, which will be filtered out to certain bandwidth using a the required filter. Then the mixer is used to translate a frequency to another. The power combiner and splitters are used to split the signal or combine them.

3.2 Filters

We have used four major filter in this experiments

- Low-pass filter : It only allows the frequency lower the cutoff frequency and rejects the higher frequency.
- High-pass filter : It only allows the frequency higher the cutoff frequency and rejects the lower frequency.
- Band-pass filter : It allows the certain specific frequency and rejects the rest

- Notch filter : It does the opposite of the band-pass filter

For low-pass and high-pass filters, f_c represents the cutoff frequency, while f_s denotes the stopband edge frequency. For band-pass and notch filters, f_1 and f_2 represent the lower and upper cutoff frequencies, respectively, and f_0 denotes the centre frequency or notch frequency.

Where P_{in} is the power measured without the filter and P_{out} is the power measured after inserting the filter. Then insertion loss is

$$IL = P_{\text{in}} - P_{\text{out}} \text{ dB}$$

3.3 Filter reflection and return loss

A part of the incident signal is reflected from the filter input, when input impedance $Z_{in} \neq Z_0$. Where Z_0 is tthe impedance of the RF system.

The voltage reflection coefficient is

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{V_{max} - V_{min}}{V_{max} + V_{min}},$$

where V_{inc} and V_{ref} are the incident and reflected voltage amplitudes, respectively.
The value of the $|\Gamma|$ says that the fraction of the incident voltage amplitude that is reflected. For a perfectly matched filter, $Z_{\text{in}} = Z_0$ and $|\Gamma| = 0$ and for complete reflection, $|\Gamma| = 1$.

The return loss is given as

$$RL = -20 \log_{10} |\Gamma| \, | \quad \text{dB}.$$

A well-matched filter has a small reflection coefficient and a large return loss. A poorly matched filter has a larger reflection coefficient and a smaller return loss.

3.4 Notch filter

A stub is a open ended transmission line which is coaxially connected in shunt through the Tee connector, behaving as a resonator. At its quarter-wave resonant frequency, the open-ended stub behaves like a short circuit at the T-junction. This gives the RF signal a very low-impedance path, so the output signal becomes very small, producing the notch.
The physical length of the stub is approximately

$$L = \frac{\lambda}{4} = \frac{VF \cdot c}{4 f_0}$$

$\forall$ L is the stub length,
λ is the guided wavelength in the cable,
VF is the velocity factor,
c is the speed of light in vacuum
f_0 is the expected notch frequency.

3.5 Mixer

An RF mixer is a three-port device that takes an RF signal and a local oscillator (LO) signal as inputs. It produces new signals as the sum and difference of their frequencies:

$$f_{IF} = f_{RF} + f_{LO}$$

and

$$f_{IF} = |f_{RF} - f_{LO}|.$$

In a receiver, a filter is used to select the required intermediate frequency (IF) from these signals.

If P_{RF} is the input RF power and P_{IF} is the measured power of the selected IF signal, the conversion loss is given by

$$\text{Conversion loss} = P_{RF} - P_{IF} \quad \text{dB}.$$

A smaller conversion loss means that more of the input RF power is converted to the required IF signal.

3.6 Power splitter/combiner

A power splitter splites the input power between two or more output ports. A practical splitter has a small additional insertion loss.

$\forall P_{\text{in}}$ be the input power
$P_{\text{out 1}}$ and $P_{\text{out 2}}$ be the powers measured at the two output ports.
The losses to the two output ports are

$$\text{Loss}_1 = P_{\text{in}} - P_{\text{out 1}}$$

and

$$\text{Loss}_2 = P_{\text{in}} - P_{\text{out 2}}$$

A power combiner works in a reverse way and the output depends on the frequencies, amplitudes, and relative phases of the two input signals.

4 Componemts required

RF signal generator(s), Spectrum analyser, Digital storage oscilloscope, Filter, Mixer, Power splitter/combiner, RF Tee connector, Open-ended coaxial stub, Short reference coaxial cable, 50Ω terminations, Suitable RF cables.

5 Experiment setup

The RF signal generator is connected to the device under test (DUT), and the output is measured using a spectrum analyser to determine filter insertion loss, mixer conversion loss, power-splitter/combiner performance, and amplifier characteristics.

For filter return-loss measurement, connect the RF signal generator to one port of the Tee connector, the oscilloscope to the second port, and the filter input to the third port. Terminate the filter output with a 50 Ω load and set the oscilloscope input impedance to 1 MΩ to minimize loading. Use the same cables and adapters throughout the measurement and perform the frequency sweep over a small, relatively flat part of the filter passband. This method gives an approximate return-loss measurement since cable loss, connector mismatch, and oscilloscope loading are not calibrated out.

6 Observations and graphs

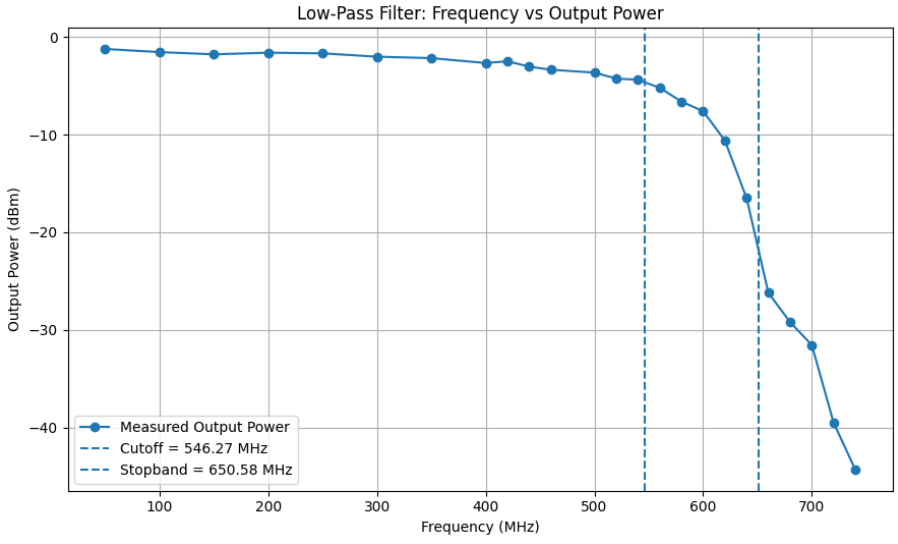
Observations of Low pass filter

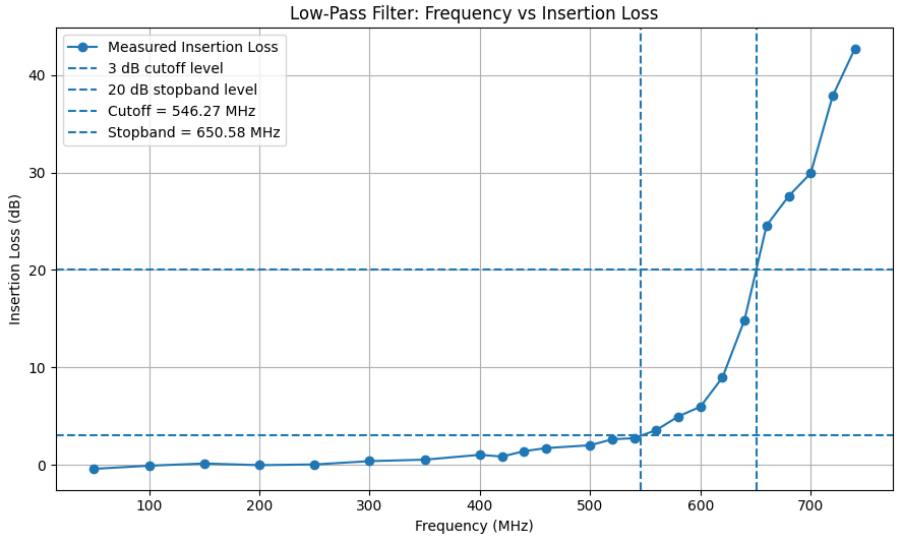
Loss due to transmission wire = 1.63 dBm Supplied power = 0 dBm

Frequency (MHz)	P_{in} (dBm)	P_{out} (dBm)	IL (dB)
50.00	-1.63	-1.22	-0.41
100.00	-1.63	-1.54	-0.09
150.00	-1.63	-1.77	0.14
200.00	-1.63	-1.60	-0.03
250.00	-1.63	-1.67	0.04
300.00	-1.63	-2.01	0.38
350.00	-1.63	-2.16	0.53
400.00	-1.63	-2.66	1.03
420.00	-1.63	-2.48	0.85
440.00	-1.63	-3.03	1.40
460.00	-1.63	-3.35	1.72
500.00	-1.63	-3.64	2.01
520.00	-1.63	-4.25	2.62
540.00	-1.63	-4.37	2.74
560.00	-1.63	-5.20	3.57
580.00	-1.63	-6.60	4.97
600.00	-1.63	-7.60	5.97
620.00	-1.63	-10.60	8.97
640.00	-1.63	-16.50	14.87
660.00	-1.63	-26.20	24.57
680.00	-1.63	-29.20	27.57
700.00	-1.63	-31.56	29.93
720.00	-1.63	-39.50	37.87
740.00	-1.63	-44.32	42.69
760.00	-1.63	-39.48	37.85

Table 1: Low-pass filter measurements.

The first 3-dB crossing gives $f_c \approx$546.3 MHz, identifying the device as a low-pass filter.

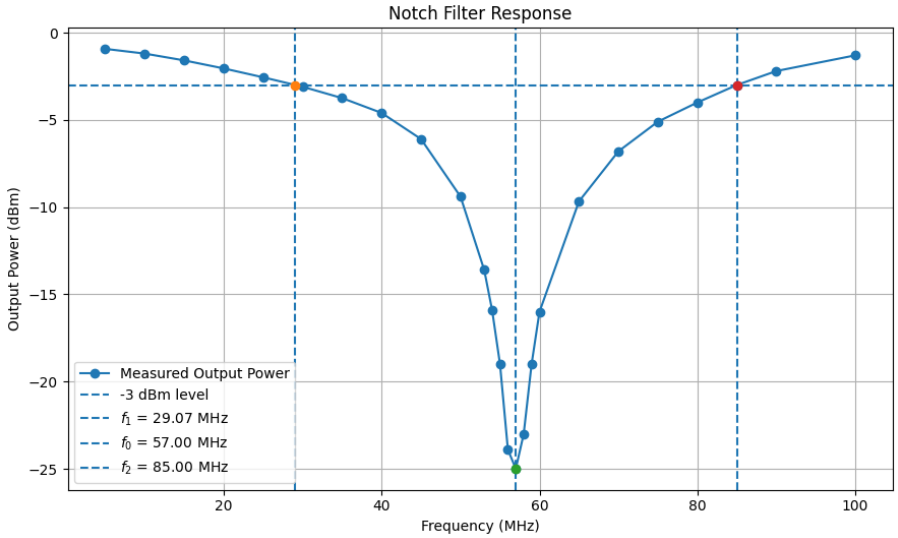




Observations of Quarter-wave stub
Length of the transmission wire = 0.925m

Frequency (MHz)	P_{in} (dBm)	P_{out} (dBm)	Attenuation (dB)
5	-1.62	-0.93	-0.69
10	-1.62	-1.20	-0.42
15	-1.62	-1.59	-0.03
20	-1.62	-2.05	0.43
25	-1.62	-2.56	0.94
30	-1.62	-3.10	1.48
35	-1.62	-3.75	2.13
40	-1.62	-4.60	2.98
45	-1.62	-6.10	4.48
50	-1.62	-9.40	7.78
53	-1.62	-13.60	11.98
54	-1.62	-15.90	14.28
55	-1.62	-19	17.38
56	-1.62	-23.90	22.28
57	-1.62	-25	23.38
58	-1.62	-23	21.38
59	-1.62	-19	17.38
60	-1.62	-16	14.38
65	-1.62	-9.67	8.05
70	-1.62	- 6.8	5.18
75	-1.62	-5.10	3.48
80	-1.62	-4	2.38
85	-1.62	-3	1.38
90	-1.62	-2.20	0.58
100	-1.62	-1.30	-0.32

Table 2: Stub measurements.



Observations of Mixer

f_{RF}	f_{LO}	Expected f_{IF}	Measured f_{IF}	P_{IF} (dBm)
30	10	20	19.99	-13.35
30	10	40	39.99	-13.37
40	10	50	49.97	-13.30
40	10	30	29.97	-13.50
50	10	60	59.99	-13.30
50	10	40	39.99	-13.20

Table 3: Mixer measurements.

Observations of Power splitter/combiner

f (MHz)	P_{in}	$P_{out,1}$	$P_{out,2}$	Loss 1	Loss 2
50.00	0.00	-4.50	-4.63	-4.50	-4.63
100.00	0.00	-4.50	-4.51	-4.50	-4.51
150.00	0.00	-4.80	-4.80	-4.80	-4.80

Table 4: Power splitter measurements.

f (MHz)	P_1	P_2	P_{out}
50	-3	-3	-7.26
100	-3	-3	-7.41
150	-3	-3	-7.54

Table 5: Power combiner measurements.

Observations of return loss

Frequency range (MHz)	$V_{max}(mV)$	$V_{min}(mV)$	$V_{ref}(mV)$	$V_{inc}(mV)$	RL (dB)
268 - 220	155.77	129.75	-13.01	142.76	20.8066

7 Calculations and results

Calculate the expected notch frequency:

$$f_0 = \frac{VF \cdot c}{4L}$$

$$\forall\ VF = 0.66 \text{ and } L = 92.5cm$$

$$f_0 = \frac{0.66 \times 3 \times 10^8}{4 \times 0.925}$$

$$f_0 = 53.51 \times 10^6 Hz$$

Calculations of return loss

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{V_{max} - V_{min}}{V_{max} + V_{min}}$$

$$\Gamma = \frac{13.01}{142.76} = 0.091131$$

$$RL = -20log_{10}(|\Gamma|)$$

$$RL = -20log_{10}(|0.091131|)$$

$$RL = 20.8066dB$$

Results

Quantity	Symbol	Measured/Calculated Value
Low-pass cutoff and stopband frequencies	f_c, f_s	$f_c = 546.3MHz,\ f_s = 740MHz$
Notch frequencies	f_1, f_0, f_2	$f_1 = 29.07MHz, f_0 = 57MHz, f_2 = 85MHz$
Filter reflection coefficient	$\|\Gamma\|$	0.091131
Filter return loss	RL	20.8066 dB
Calculated stub notch frequency	f_0	53.51 MHz
Power-splitter output powers	$P_{\text{out 1}}, P_{\text{out 2}}$	$P_{\text{out 1}} = -4.5dBm, P_{\text{out 2}}$ = -4.5dBM

The filter shows clear low-pass behavior with a cutoff near 546.3 MHz. The open stub gives a strong notch near 57 MHz. Mixer products agree closely with the expected sum/difference frequencies. The splitter shows greater-than-ideal 3-dB loss, consistent with practical losses.

8 Conclusion

The RF components were successfully characterized using the supplied observation data. The measured low-pass cutoff is approximately 546.3 MHz, the stub notch is approximately 57 MHz, splitter losses are approximately 4.5–4.8 dB.

- The filter shows clear low-pass behavior with a cutoff near 546.3 MHz.

- Insertion loss is used to characterize how much signal power is lost when a component is inserted into the RF path.
- A quarter-wave open-circuit stub can act as a notch filter because, at resonance, it provides a low-impedance path that strongly suppresses the signal at the output.
- An RF mixer performs frequency translation, by summing and difference of frequency components.
- Practical power splitter or power combiner have additional insertion loss compared with the ideal case.
- Return loss and reflection coefficient are important for assessing impedance matching. Good matching corresponds to low reflection and high return loss.

References

[1] SPA617 Radio Astronomy Laboratory, *RF Components: Characterization and Measurement*, Revision 2.0, August 4, 2026.

AI usage : Used chatgpt to plot the graphs and formatting the latex file.